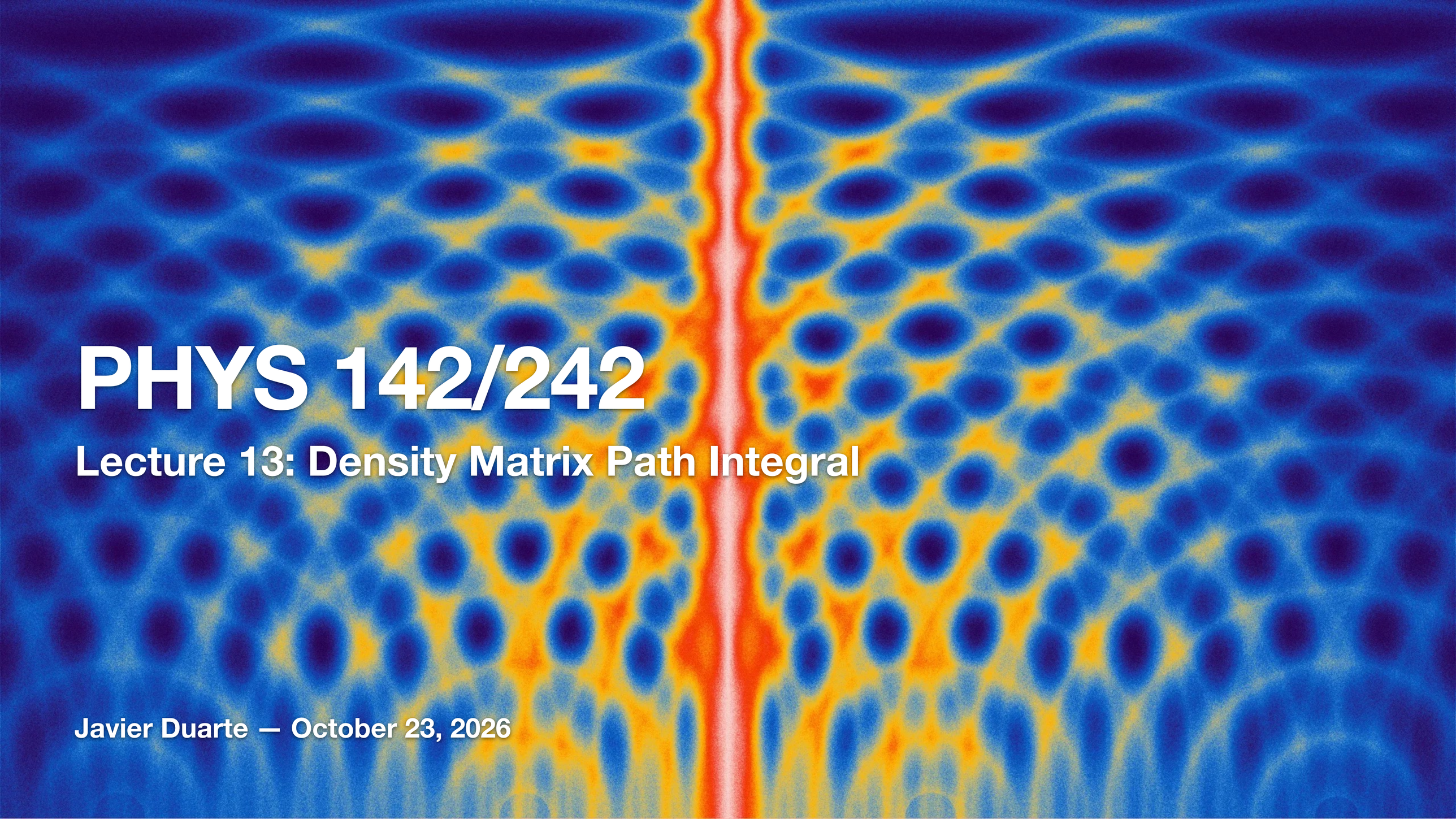


PHYS 142/242

Lecture 13: Density Matrix Path Integral

Javier Duarte — October 23, 2026

Canonical ensemble

In the canonical ensemble, the probability p_n that the system is in a state of energy E_n is

$$p_n = \frac{1}{Z} e^{-E_n/(k_\mathrm{B}T)},$$

where $k_\mathrm{B} = 1.38 \times 10^{-23}\,\mathrm{J/K}$ is the Boltzmann constant and

$$Z = \sum_n e^{-E_n/(k_\mathrm{B}T)}$$

is the partition function.

Convenient to define “inverse temperature” $\beta = 1/(k_\mathrm{B}T)$

Partition function relations

The partition function can be used to compute all other standard thermodynamic quantities, including

Helmholtz free energy (measures the useful work obtainable from a closed system at constant temperature) $F = -k_{\mathrm{B}} T \ln Z$

Internal energy $U = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}$

Entropy $S = -\frac{\partial F}{\partial T}$

Density matrix reminder

Density matrix is a generalization of usual state vectors or wavefunctions that can only represent pure states

Density matrices can represent mixed ensembles (also called mixed states)

Mixed ensembles arise in quantum mechanics in two different situations:

1. A system is prepared with numerous pure states in an ensemble
2. A system is entangled with another and the combined state is unknown

Density matrix reminder

Consider Hilbert space with two states $|0\rangle, |1\rangle$, then the ***density matrix*** is

$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j| = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{01}^* & \rho_{11} \end{pmatrix}$ where some pure states $|\psi_j\rangle$ (not necessarily orthogonal) are prepared with probability p_j each

Properties:

- positive semi-definite
- self-adjoint (Hermitian) $\rho^\dagger = \rho$
- $\mathrm{tr}(\rho) = 1$

Pure states vs. mixed states

An ensemble is prepared to have half of its systems in state $|0\rangle$ and the other half in $|1\rangle$ (***mixed state***), then

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

On the other hand, imagine the entire system is the superposition (***pure state***) $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, then

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Note: If system is in a pure state then

- $\rho = |\psi\rangle\langle\psi|$
- ρ is idempotent $\rho^2 = \rho$
- $\mathrm{tr}(\rho^2) = 1$

Density matrix

The (unnormalized) density matrix is defined as

$$\tilde{\rho}(x', x) = \sum_n \phi_n(x')\phi_n^*(x)e^{-\beta E_n}$$

We see that

$$Z = \int \tilde{\rho}(x, x)dx \equiv \mathrm{Tr}(\tilde{\rho}) = \sum_n e^{-\beta E_n}$$

Note: in most textbooks, we usually call the $\rho(x', x) = \tilde{\rho}(x', x)/Z$ the proper normalized density matrix

Relation to path integral

This formulation bears a striking resemblance to the path integral

$$K(x', t; x, 0) = \sum_n \phi_n(x')\phi_n^*(x)e^{-(i/\hbar)E_n t},\ t > 0$$

whose trace gives the spectrum

$$\mathrm{Tr}(K) = \sum_{n=1}^{\infty} e^{-(i/\hbar)E_n t}$$

If we set $it = \hbar\beta$ (i.e., $t = -i\hbar\beta$), the two are ***formally identical***

This is called ***Wick rotation*** or ***analytic continuation to imaginary time***

Density matrix equation

We can show that the density matrix satisfies this differential equation

$$\frac{\partial \tilde{\rho}(b,a)}{\partial \beta} = -\hat{H}_b \tilde{\rho}(b,a)$$

Similar to the Schrödinger equation for the propagator K

$$\frac{\partial K(b,a)}{\partial t_b} = -\frac{i}{\hbar}\hat{H}_b K(b,a) \quad \text{for} \quad t_b > t_a$$

Both are Green's functions for their respective differential equations.

We can solve them in the same way

Density matrix solution

The solution for the propagator can be built up over “zig-zag paths” with $x_a = x_0$, $x_b = x_N$, $t_a = 0$, and $t_b = N\epsilon$

$$K(b,a) = \lim_{\epsilon \to 0} \int \cdots \int dx_1 \cdots dx_{N-1} \left(\frac{2\pi i \hbar \epsilon}{m} \right)^{-\frac{N}{2}} \exp \left[\frac{i}{\hbar} \left(\sum_{i=1}^{N} \frac{m(x_i - x_{i-1})^2}{2\epsilon} - \epsilon V \left(\frac{x_{i-1} + x_i}{2} \right) \right) \right]$$

Performing our replacement $it = \hbar\beta$ and $i\epsilon = \eta$

$$\tilde{\rho}(b,a) = \lim_{\eta \to 0} \int \cdots \int dx_1 \cdots dx_{N-1} \left(\frac{2\pi \hbar \eta}{m} \right)^{-\frac{N}{2}} \exp \left[-\frac{1}{\hbar} \left(\sum_{i=1}^{N} \frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V \left(\frac{x_{i-1} + x_i}{2} \right) \right) \right]$$

You can treat this as just a “formal trick” to solve for the density matrix

Interpretation

$$\tilde{\rho}(b,a) = \lim_{\eta\to 0} \int \cdots \int dx_1 \cdots dx_{N-1} \left(\frac{2\pi\hbar\eta}{m}\right)^{-\frac{N}{2}} \exp\left[-\frac{1}{\hbar}\left(\sum_{i=1}^{N} \frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V\left(\frac{x_{i-1}+x_i}{2}\right)\right)\right]$$

What does it mean? Although we treat $\beta\hbar$ as an “imaginary time” it is actually related to the inverse of temperature

This equation says that the density matrix for a temperature $T = (k_{\mathrm{B}}\beta)^{-1}$ is given by considering all the possible paths, or "motions," by which the system can travel between the initial and final configurations in the “time" $\beta\hbar$

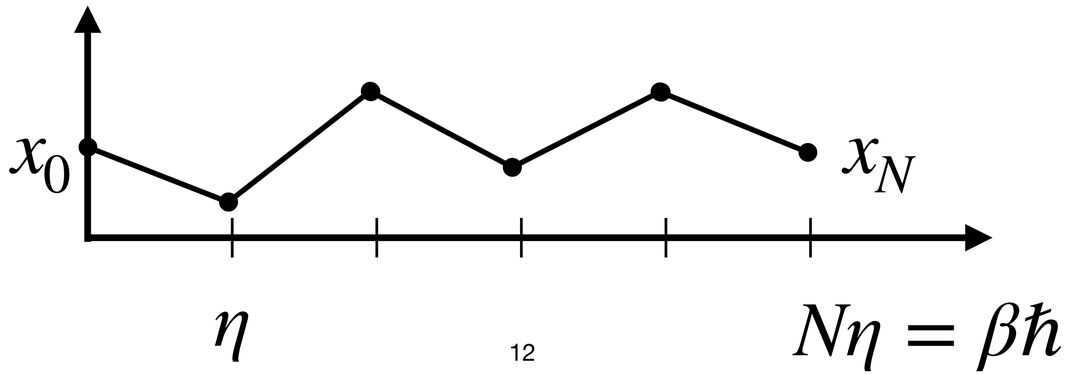
Partition function

We're usually most interested in the partition function

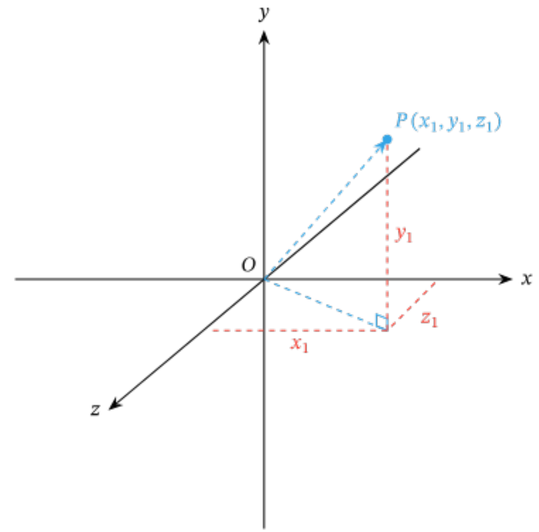
$$Z = \int dx_a \; \tilde{\rho}(a, a)$$

That is we only consider paths where the initial configuration is the same as the final configuration $x_N \equiv x_0$, and then we integrate over all such configurations ($\eta \equiv \delta\tau$)

$$Z = \lim_{\eta \to 0} \int \cdots \int dx_0 \cdots dx_{N-1} \left(\frac{2\pi\hbar\eta}{m} \right)^{-\frac{N}{2}} \exp \left[-\frac{1}{\hbar} \left(\sum_{i=1}^{N} \frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V \left(\frac{x_{i-1} + x_i}{2} \right) \right) \right]$$



Probability



Each vector $(x_0, x_1, \ldots, x_{N-1})$ in N-dimensional space

represents a zig-zag path (with $x_N \equiv x_0$)

The probability density over this N-dimensional space is

$$p(x_0, \ldots, x_{N-1}) = \frac{1}{Z}\left(\frac{2\pi\hbar\eta}{m}\right)^{-\frac{N}{2}} \exp\left[-\frac{1}{\hbar}\left(\sum_{i=1}^{N}\frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V\left(\frac{x_{i-1}+x_i}{2}\right)\right)\right]$$

$$= \frac{\exp\left[-\frac{1}{\hbar}\left(\sum_{i=1}^{N}\frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V\left(\frac{x_{i-1}+x_i}{2}\right)\right)\right]}{\int\cdots\int dx_0\cdots dx_{N-1}\exp\left[-\frac{1}{\hbar}\left(\sum_{i=1}^{N}\frac{m(x_i - x_{i-1})^2}{2\eta} + \eta V\left(\frac{x_{i-1}+x_i}{2}\right)\right)\right]}$$

Connection with heat bath

Reminder, for an electron in heat bath at temperature T, the probability to be found in the nth energy eigenstate is

$$p_n = \frac{e^{-E_n/(k_\mathrm{B}T)}}{Z} \text{ with } Z = \sum_n e^{-E_n/(k_\mathrm{B}T)}$$

So the probability for the electron to be found in a given position x is

$$\rho(x,x) = \sum_n p_n \left| \phi_n(x) \right|^2$$

Low temperature limit

Note, as we approach $T \to 0$, i.e. $\beta \to \infty$

$Z \sim e^{-E_0/(k_B T)}$ (i.e. only the first eigenvalue dominates the sum)

and

$\rho(x, x) \sim p_0 \, |\, \phi_0(x) \,|^2$ with $p_0 \sim 1$

So if we set the temperature low enough, our probability distribution should converge to the ground state

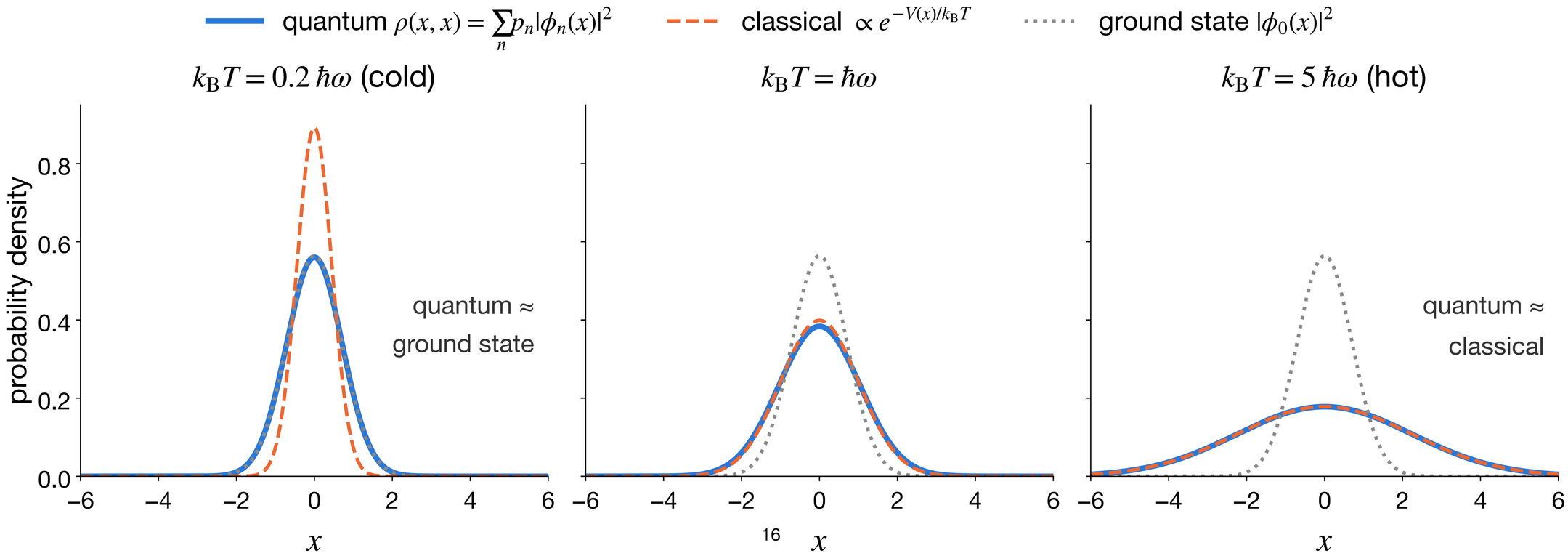
We can use the path integral (with imaginary time) to directly calculate the ground state

High temperature limit

Cold: the quantum distribution settles into the ground state, while the classical one keeps shrinking toward $x = 0$

Hot: the quantum curve becomes the classical Boltzmann distribution.

Crossover: happens around $k_B T \approx \hbar\omega$



Monte Carlo

Use Monte Carlo methods to sample $p(x_0, \ldots, x_{N-1})$ probability distribution

Intractable to sample directly using standard methods like hit-and-miss, because of the curse of dimensionality

Instead, we will use *importance sampling* to focus on areas of high probability and introduce the method of ***Markov chain Monte Carlo***

